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**A PROJECT REPORT
ON**

Hostel Buddy (IIITU Hostel website)

Bachelor of Technology

in

Information Technology

By

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SCHOOL OF COMPUTING

**INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
UNA HIMACHAL PRADESH**

DECEMBER 2023

BONAFIDE CERTIFICATE

This is to certify that the project titled Hostel Buddy (IIITU Hostel website) is a bonafide record of the work done by

AKSHIT MEHTA (22307)

in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Information Technology of the INDIAN INSTITUTE OF INFORMATION TECHNOLOGY UNA, HIMACHAL PRADESH, during the year 2023 - 2024.

under the guidance of

Ms. HEM LATA

Project viva-voce held on: _____

Internal Examiner

External Examiner

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ABSTRACT

"Hostel Buddy" introduces an innovative College Hostel Website designed to streamline and enhance the efficiency of hostel operations within academic institutions. The project addresses common challenges faced by colleges and universities in managing on-campus hostels. The existing manual processes, particularly for tasks like online mess complaints, civil complaints, and leave applications, are often cumbersome, error-prone, and lacking effective communication channels.

The proposed solution integrates a user-friendly platform that facilitates online mess complaints, civil complaints, and leave applications, providing a centralized and efficient system. Through Hostel Buddy, students can easily submit their concerns and requests, while administrators gain access to a systematic management interface. This initiative aims to eliminate operational inefficiencies, reduce errors, and enhance overall communication between students and hostel administrators.

In summary, Hostel Buddy offers a comprehensive College Hostel Management System to address the complexities of hostel administration, promoting efficiency and improved user experience for both students and administrators.

Keywords: HTML, CSS, JAVASCRIPT, FIREBASE

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AKSHIT MEHTA (22307)

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LIST OF ACRONYMS

HTML - Hyper Text Markup Language.

CSS - Cascading Style sheet.

JS - JavaScript.

DOM – Document Object Model.

DB – Data Base.

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Chapter 1

Introduction

1.1 Brief description of the application:

The college hostel management system is a comprehensive software application designed to streamline and automate various tasks related to hostel administration. The primary features of the system include:

1.1.1 Online Mess Complaints:

- Users, typically hostel residents, can submit mess-related complaints through the online platform.
- The system records and organizes these complaints for prompt resolution by the hostel staff or management.
- Notifications may be sent to relevant staff members to ensure timely response and resolution.

1.1.2 Civil Complaints:

- Residents can file civil complaints regarding issues within the hostel premises, such as maintenance problems, electrical issues, plumbing concerns, etc.
- The system logs and categorizes these complaints for the maintenance or administrative team to address.
- Automated alerts may be generated to notify the responsible parties about new complaints.

1.1.3 Leave Applications:

- Hostel residents can submit leave applications through the system, specifying the dates and reasons for their intended absence.
- The application tracks leave requests and their status, providing a transparent process for both residents and hostel authorities.
- Notifications are sent to relevant personnel for leave approval or denial.

1.1.4 User Authentication and Access Control:

- The system incorporates secure user authentication to ensure that only authorized individuals can access and modify specific information.
- Different levels of access may be granted to hostel staff, administrators, and residents based on their roles and responsibilities.

1.1.5 Data Management:

- The application maintains a centralized database to store and manage information related to hostel residents, complaints, leave records, and other relevant data.
- Regular backups and data integrity checks are implemented to safeguard information.

1.2 Problem Definition:

The problem definition for the college hostel management system arises from the challenges and inefficiencies in traditional manual methods of handling hostel-related tasks. Here are some key problems that the application seeks to address:

1.2.1 Manual Complaint Handling :

- Traditional methods of handling mess and civil complaints involve manual paperwork, making it time-consuming and prone to errors.
- Lack of a centralized system can lead to delays in complaint resolution and difficulty in tracking the status of various complaints.

1.2.2 Inefficient Leave Management:

- Managing leave applications manually can result in a cumbersome process for both residents and hostel authorities.
- There may be a lack of transparency in leave approval processes, leading to confusion and potential conflicts.

1.2.3 Communication Challenges:

- Traditional communication methods may be ineffective in disseminating important information to hostel residents.
- Without a centralized platform, announcements and updates may be missed or not reach all residents in a timely manner.

1.2.4 Data Discrepancies and Security Concerns:

- Maintaining records manually increases the risk of data discrepancies and inaccuracies.
- Security concerns arise when sensitive information about residents and their activities is not adequately protected.

1.2.5 Lack of Insights and Analytics:

- Without proper tools for reporting and analytics, hostel management may lack the necessary insights into trends, patterns, and areas that require improvement.

1.2.6 Time-Consuming Administrative Tasks:

- Administrative tasks, such as managing complaints, leave applications, and resident data, can be time-consuming when handled manually.
- Automation can significantly reduce the workload on hostel staff and improve overall efficiency.

1.2.7 Difficulty in Tracking Trends:

- Traditional systems may lack the capability to track trends and recurring issues.

1.3 Objectives:

1.3.1 Hostel Management Software:

- Implementation of a Centralized System.
- Deploy a robust hostel management system that centralizes all data and processes.
- Ensure the system is scalable and can accommodate future needs.
- Paper-Free Processes:
- Online Complaint Resolution
- Allow residents to submit complaints through the online portal.
- Implement a ticketing system for tracking and resolving complaints.
- Digital Leave Applications:
- Enable residents to apply for leave online with an approval workflow.

1.3.2 Communication Platform:

- Integrated Messaging System:
- Include an internal messaging system for residents and staff.
- Ensure notifications are sent via email .
- Centralize announcements and notices within the system.
- Implement push notifications for urgent communications.

1.3.3 Mobile Accessibility:

- Responsive Design.
- Ensure the system is accessible on mobile devices.
- Develop a dedicated mobile app for easy access.

1.3.4 Data Security and Privacy:

- Secure Access Controls:
- Implement role-based access controls to protect sensitive information.
- Encrypt data to ensure privacy.

1.3.5 Training and Support:

- User Training:
- Conduct training sessions for residents and staff on using the new system.
- Provide ongoing support through helpdesk services.

1.3.6 Sustainability Measures:

- Encourage digital documentation and communication.
- Implement recycling measures for any necessary physical documents.

- Ensure servers and infrastructure are energy-efficient.
- Promote eco-friendly practices in the hostel.

1.3.7 Integration with Existing Systems:

- Ensure smooth integration with existing IIIT UNA systems.
- Avoid disruption to ongoing operations during the transition.

Chapter 2

Review of Literature

2.1 HTML

The HyperText Markup Language or HTML is the standard markup language for documents designed to be displayed in a web browser. HTML elements are the building blocks of HTML pages. With HTML constructs, images and other objects such as interactive forms may be embedded into the rendered page. HTML provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes, and other items. HTML was first created by Tim Berners-Lee, Robert Cailliau, and others starting in 1989. It stands for Hyper Text Markup Language. Hypertext means that the document contains links that allow the reader to jump to other places in the document or to another document altogether.

The latest version is known as HTML5.

2.2 CSS

CSS is designed to enable the separation of content and presentation, including layout, colors, and fonts. This separation can improve content accessibility; provide more flexibility and control in the specification of presentation characteristics; enable multiple web pages to share formatting by specifying the relevant CSS in a separate .css file, which reduces complexity and repetition in the structural content; and enable the .css file to be cached to improve the page load speed between the pages that share the file and its formatting.

2.3 JS

JavaScript (js) is a light-weight object-oriented programming language which is used by several websites for scripting the webpages. It is an interpreted, full-fledged programming language that enables dynamic interactivity on websites when applied to an HTML document.

2.4 FIRE BASE

Firebase is a mobile and web application development platform by Google. It offers a real-time NoSQL database, authentication, cloud functions, and hosting services, providing developers with a scalable and integrated solution for building feature-rich applications.



Figure 2.1: Tools logo

Chapter 3

Proposed Work

This web development-based project is a college hostel management system that empowers users to efficiently manage tasks such as online mess complaints, civil complaints, and leave applications.

3.1 Login page of my website

The login page of an hostel website serves as a critical component for user authentication and access control, extending beyond the basic verification of user credentials. Its primary function lies in securely confirming entered information, typically a username and password, against stored user data. Providing clear user feedback is essential, with loading indicators conveying ongoing authentication processes and explicit messages differentiating between successful logins and failed attempts. For new users, the login page may offer an option to register and create an account, contributing to a seamless onboarding process.

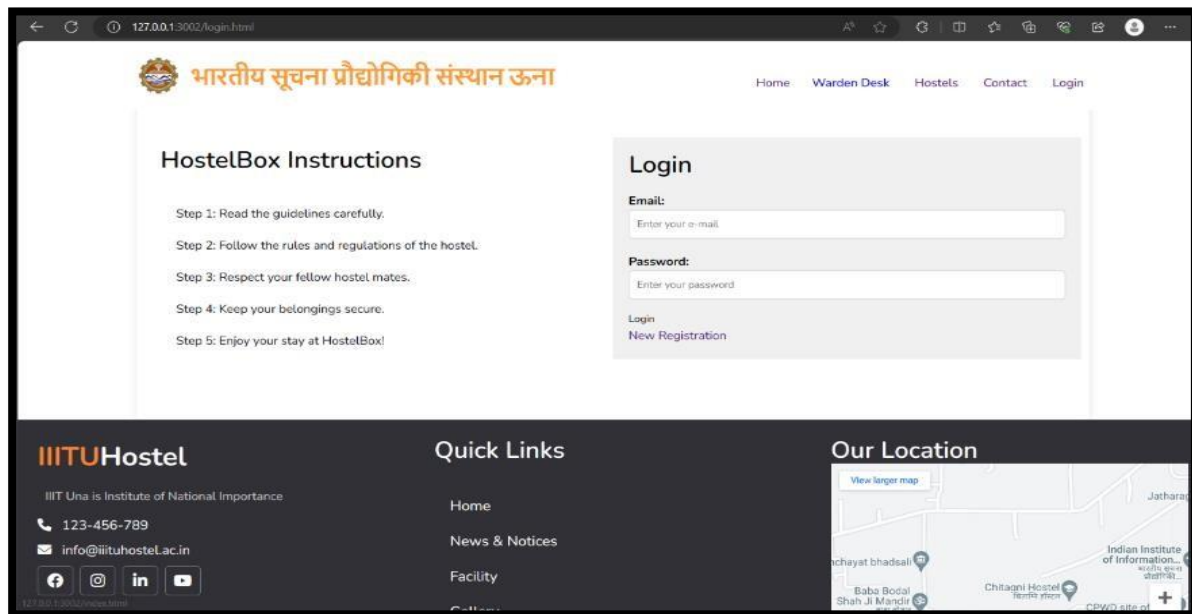


Fig 3.1: login page

3.2 Home page

Welcome to our Home Page, your central hub for hostel management. Our Warden Desk ensures swift contact for emergency purposes, providing a direct link to reach our dedicated warden. Explore the Hostels section to preview our accommodation facilities. Find essential contact information for the warden, promoting seamless communication. Utilize the Login Page to access your email ID securely. We prioritize convenience and connectivity to enhance your overall experience with our hostel management system.

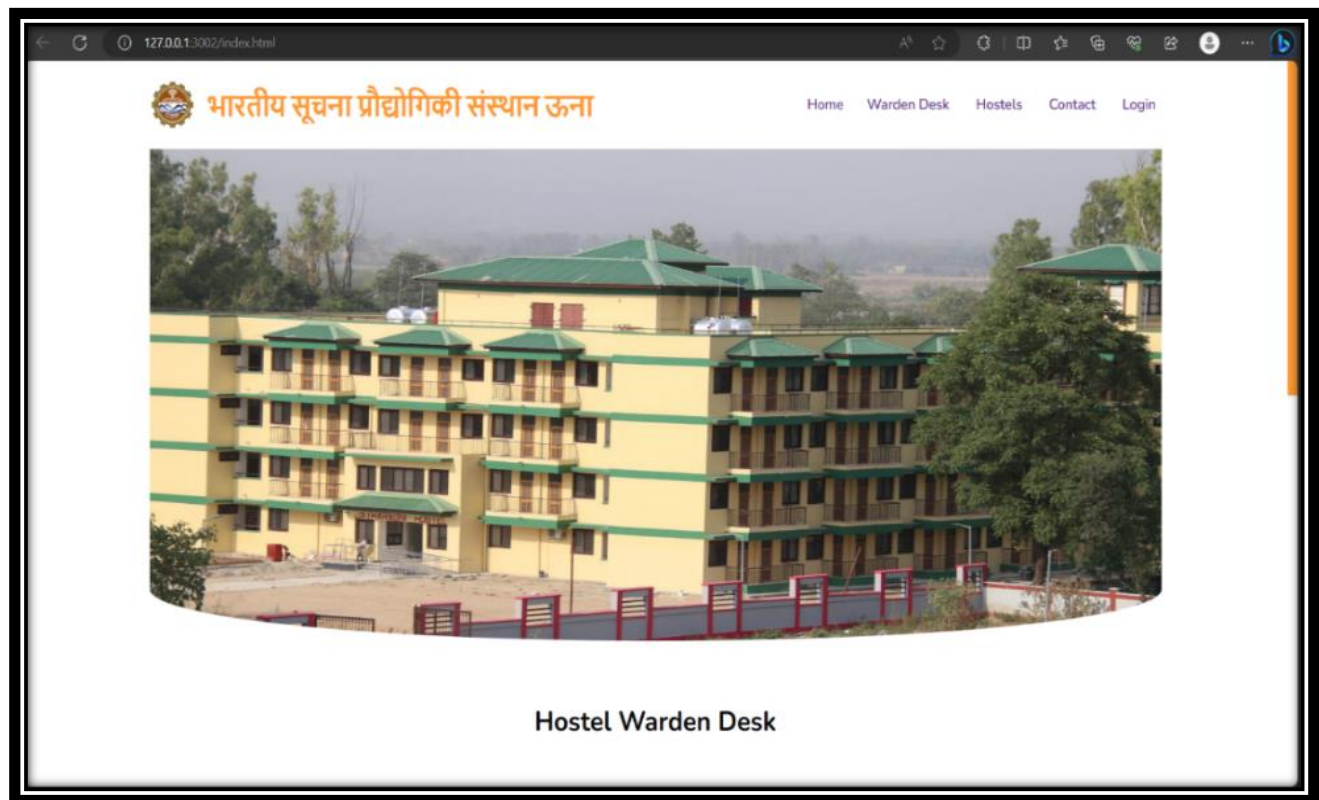


Fig 3.2: home page

3.3 List of all wardens

Welcome to the Hostel Warden Directory! Our dedicated team of wardens is available 24/7 to assist you with any emergencies or queries. Browse the list to find contact information for each warden, ensuring prompt and reliable support whenever you need it. Your safety and well-being are our top priorities, and our committed wardens are here to provide assistance, around the clock. Feel secure knowing that help is just a call away.

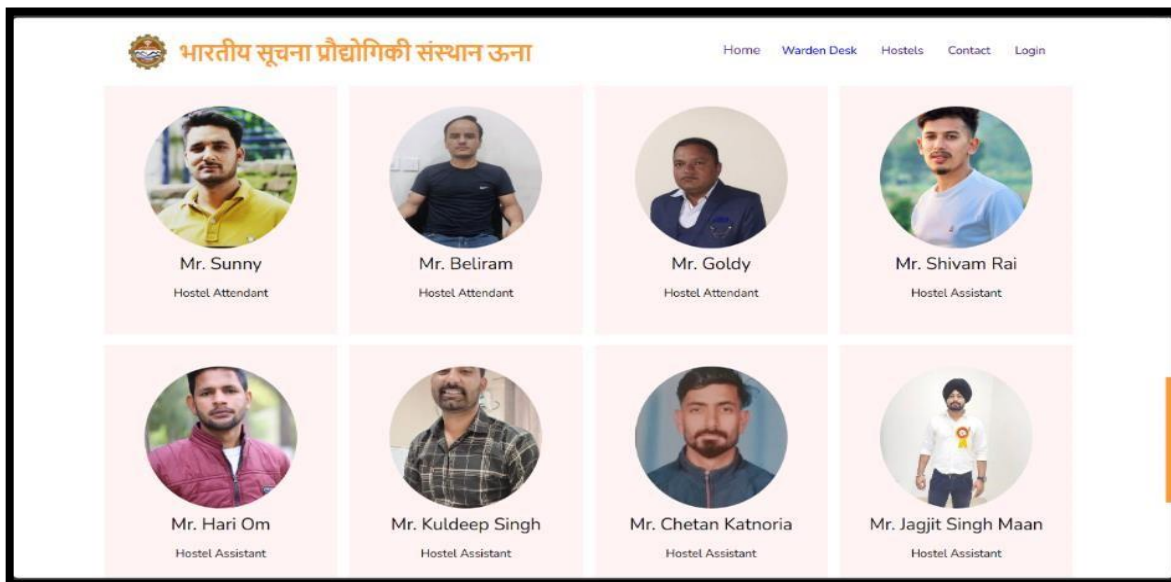


Fig.3.3: warden list

3.4 Mess Complaint Form

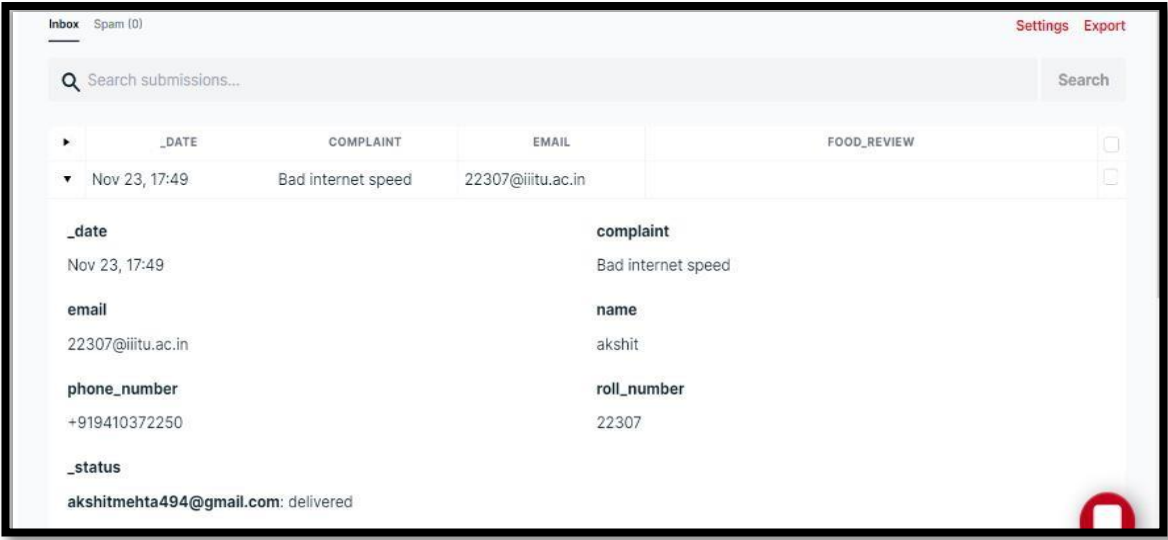
Welcome to our "Mess Complaint and Food Review Form" – an interactive platform designed to gather valuable feedback from our hostel residents. This user-friendly form allows you to express your thoughts and concerns about the hostel mess, ensuring a direct line of communication between you and our administrative team. Whether you have specific complaints or want to share your food experiences, this form is your opportunity to contribute to the enhancement of our hostel services.

The screenshot shows a digital form titled "Mess Complaint and Food Review Form" overlaid on a background image of a hostel mess hall. The form contains the following fields: Name (with placeholder "Your Name"), Roll No. (with placeholder "Your Roll Number"), Phone Number (with placeholder "Your Phone Number"), Email (with placeholder "Your Email"), Complaint (with placeholder "Enter Your Mess Complaint"), and Food Review (with placeholder "Provide Your Food Review and Suggest Changes"). A blue button labeled "Submit Complaint and Food Review" is located at the bottom of the form.

Fig.3.6: Mess complaint form

3.5 Data of Mess Complaint form

Our "Mess Complaint and Food Review Form" ensures efficient communication by forwarding key details to the hostel warden via email. This includes the submission date, student's name, roll number, contact information, and the nature of the complaint or food review. This streamlined process enables timely attention to concerns and facilitates prompt resolution. Your feedback matters, and our commitment to addressing issues promptly is central to ensuring a positive living experience in our hostel.



_DATE	COMPLAINT	EMAIL	FOOD_REVIEW
Nov 23, 17:49	Bad internet speed	22307@iiitu.ac.in	

_date Nov 23, 17:49	complaint Bad internet speed
email 22307@iiitu.ac.in	name akshit
phone_number +919410372250	roll_number 22307
_status akshitmehta494@gmail.com: delivered	

Fig.3.7: Data of mess complaint

Chapter 4

Result

The college hostel management system streamlines administrative tasks by providing an efficient platform for handling online mess complaints, civil complaints, and leave applications. This system enhances user experience, allowing residents to submit their concerns seamlessly. Through its user-friendly interface, it promotes effective communication and facilitates the swift resolution of various issues, contributing to a well-organized and responsive hostel management environment. Following are the snapshots of my website:

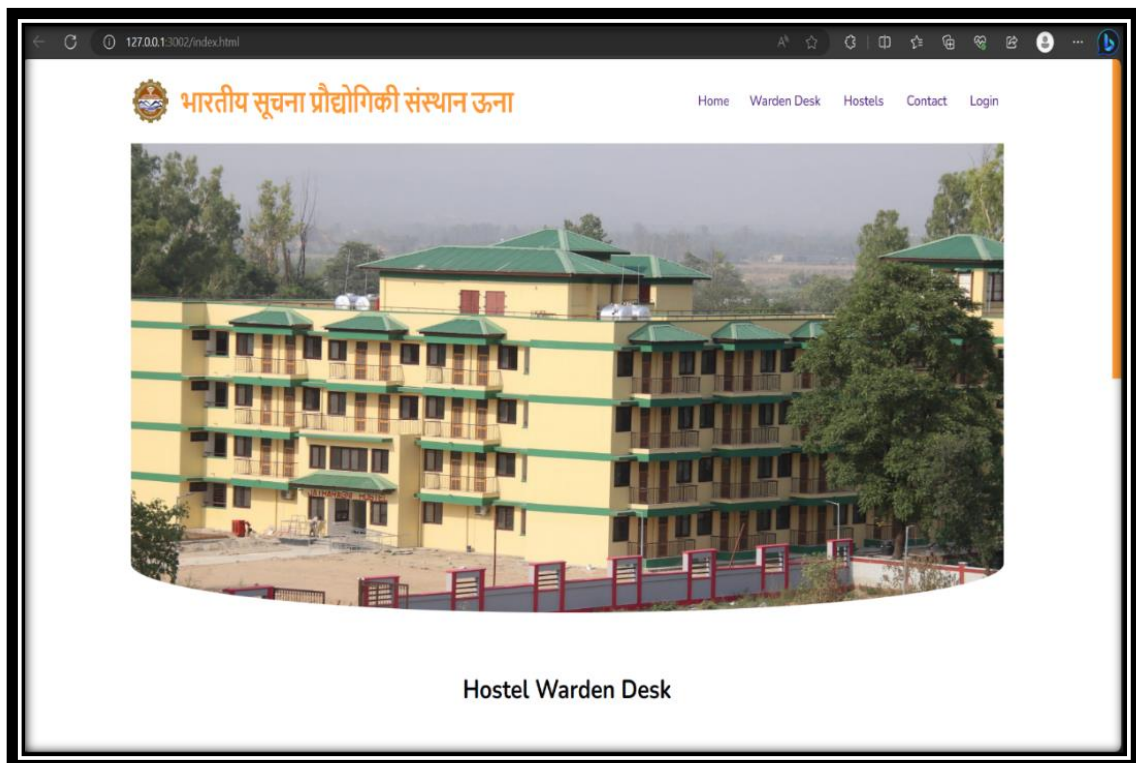


Figure 1: hostel home page

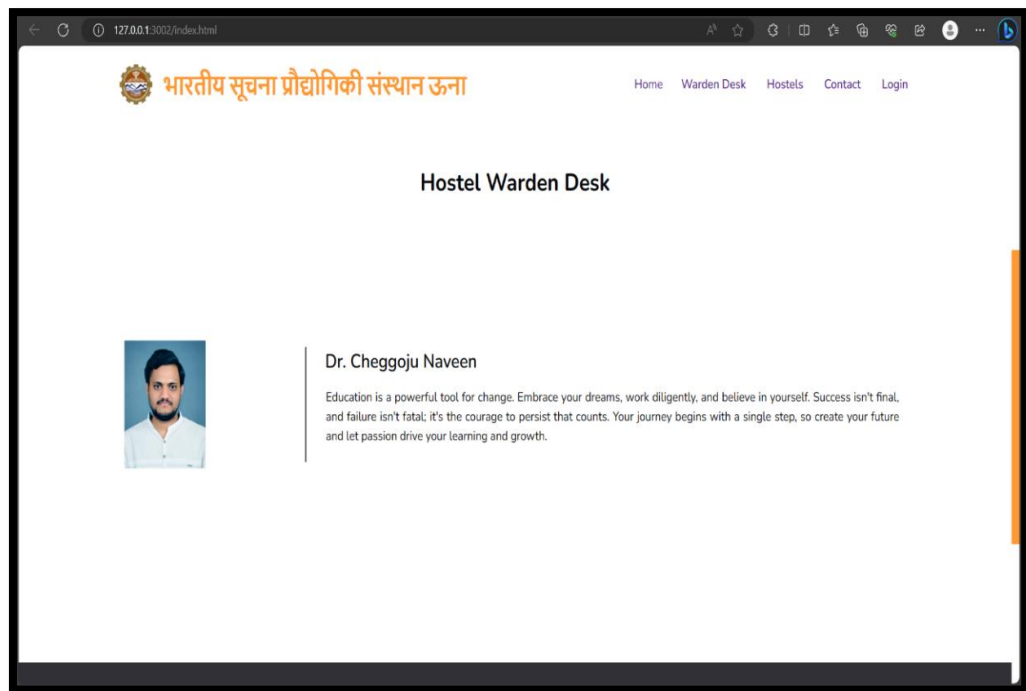


Figure 2:warden desk

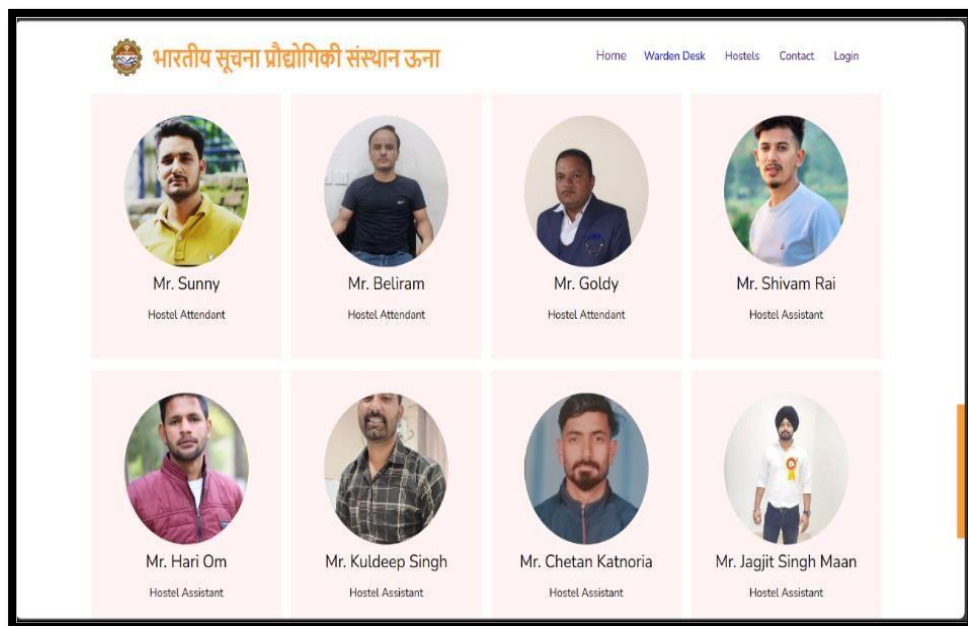


Figure 3:hostel warden

Mess Complaint and Food Review Form

Name:

Roll No:

Phone Number:

Email:

Complaint:

Food Review:

[Submit Complaint and Food Review](#)

Figure 4: Mess complaint form

Inbox Spam (0) Settings Export

Q Search submissions... Search

▶	_DATE	COMPLAINT	EMAIL	FOOD_REVIEW	☐
▼	Nov 23, 17:49	Bad internet speed	22307@iiitu.ac.in		☐

_date

Nov 23, 17:49

email

22307@iiitu.ac.in

phone_number

+919410372250

_status

akshitmehta494@gmail.com: delivered

complaint

Bad internet speed

name

akshit

roll_number

22307

Figure 5: mess complaint data in your email

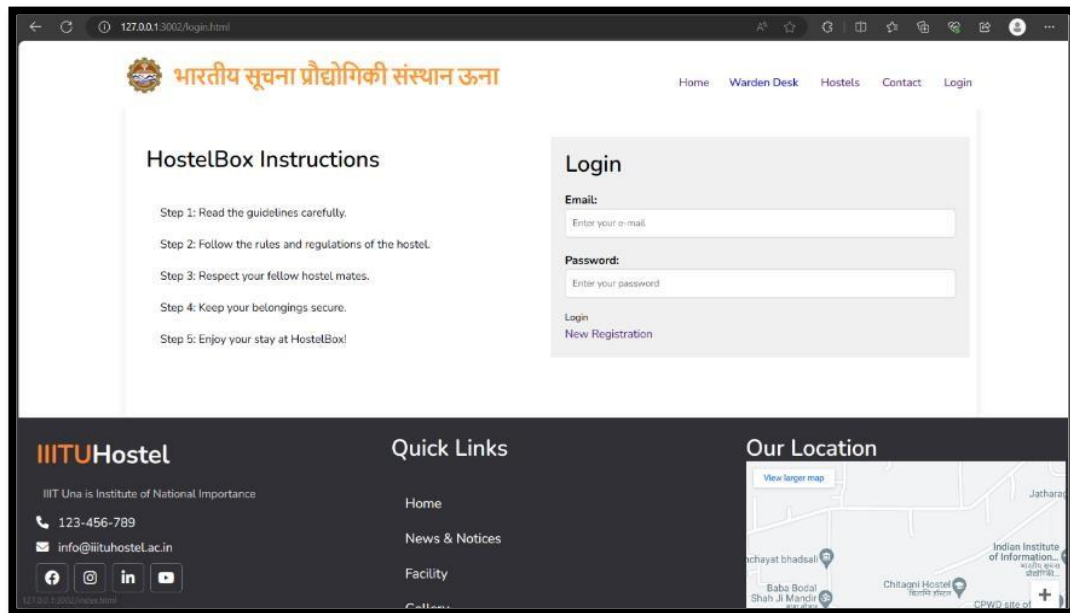


Figure 6: login page

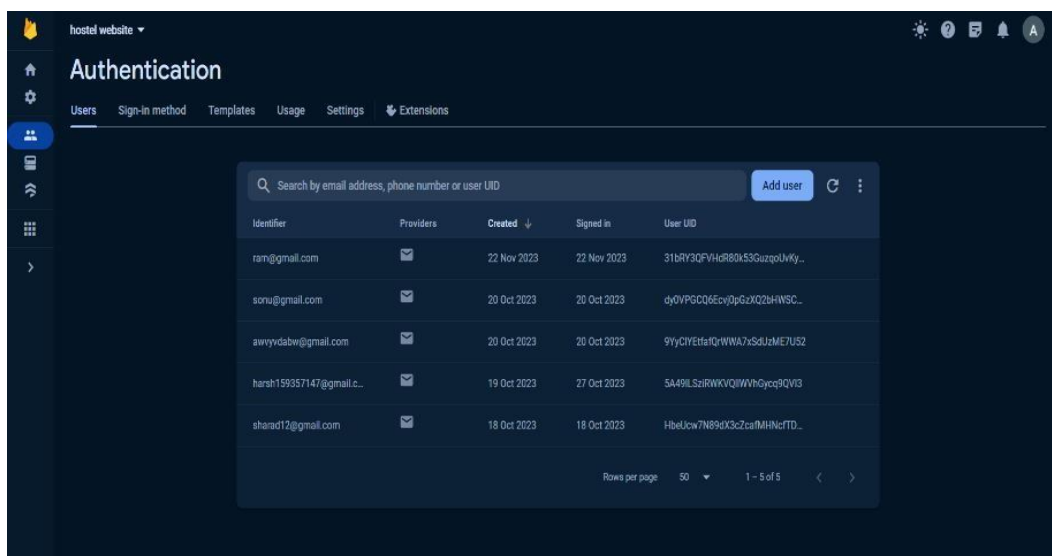


Figure 7: login Information

Chapter 4

Conclusion and Future Work

4.1 Conclusion

This website serves as a dynamic and user-centric platform, addressing the diverse needs of hostel life. From streamlined complaint handling to comprehensive hostel information, we strive to enhance user experience. The commitment to efficiency, communication, and continual improvement reflects our dedication to creating a positive and supportive living environment for all residents. Thank you for being a part of our community, and we look forward to continually evolving to meet your needs.

4.2 Future Scope

- Add NLP in the search box and add a question box so that if a question is already asked then it can be notified to the particular user.
- Add filters to avoid spam/explicit questions and answers.

References

- <https://developer.mozilla.org/en-US/>
- <https://www.w3schools.com/>
- Udemy Web development course by Angela YU. <https://www.udemy.com/>
- <https://stackoverflow.com/>
- <https://www.youtube.com/@NetNinja>

Appendix A

Code Attachments

The following is the partial / subset of the code. Code of some module(s) have been wilfully suppressed.

A.1 Sample Code

HTML Code

(Login Page)

```
<div class="login-form">
<h2>Login</h2>

<div class="form-group">
<label for="username">Email:</label>

<input type="text" id="email" name="email" placeholder="Enter your
email">
</div>

<div class="form-group">
<label for="password">Password:</label>

<input type="password" id="password" name="password" placeholder="Enter
your password">
</div>
```

```

<button type="submit" id="login" name="login"
value="Login">Login</button>
<br>

<a href="registration.html">New Registration</a>

</div>

```

(Home Page)

```

<body>

<header class="header" id="header">
<section class="wrapper nav">
<a href="./index.html" class="brand">
<p>भारतीय सूचना प्रौद्योगिकी संस्थान ऊना</p>
</a>

<div class="burger" id="burger">
<span class="burger-line"></span>

<span class="burger-line"></span>

<span class="burger-line"></span>

</div>

<span class="overlay"></span>

<nav class="navbar" id="navbar">
<ul class="menu" id="menu">
<li class="menu-item"><a href="index.html">Home</a></li>
<li class="menu-item menu-dropdown">
<span class="menu-link" data-toggle="submenu"><a
href="#director">Warden Desk</a><i class="bx bx-chevron-down"></i></span>
</li>

```

```

<li class="menu-item menu-dropdown">
<span class="menu-link" data-toggle="submenu"><a
href="hostel.html">Hostels</a><i class="bx bx-chevron-down"></i></span>
</li>

<li class="menu-item menu-dropdown">
<span class="menu-link" data-toggle="submenu"><a
href="contact.html">Contact</a><i class="bx bx-chevron-down"></i></span>
</li>

<li class="menu-item menu-dropdown">
<span class="menu-link" data-toggle="submenu"><a
href="login.html">Login</a><i class="bx bx-chevron-down"></i></span>
</li>

</ul>

</nav>

</section>

</header>

<section>

<div class="paralex-body">
<div class="paralex-head paralex-images">
<div class="paralex-text"
  </div>

  </div>
</div>

</section>

<section id="director">

<div class="directorheading">
<h1 class="director-heading">Hostel Warden Desk</h1>

```

```
</div>
```

CSS Code

```
h1 {  font-size:
36px;  font-
size: 700;
} h2 {  font-size:
32px;  font-
weight: 600;
line-height: 36px;
} h3 {  font-size:
26px;  font-
weight: 500;
line-height: 30px;
} h4 {  font-size:
24px;  font-
weight: 400;
margin: 10px 5px;
line-height: 26px;
} p {  font-size:
16px;  margin:
15px 10px;  line-
height: 25px;
} a,
a:active,
a:focus,
button {
text-
decoratio
n: none;
outline:
0;

} ul
{

list-style: none;
}
```

```
::selection {
    background-color: #f938;
}
```

```
.container {    box-
sizing: border-box;
margin: auto;    max-
width: 100%;
}
```

```
.section-title {    text-
align: center;
position: relative;
font-size: 34px;    font-
weight: 600;    padding:
20px 10px 0px;
}
```

JS Code

(Frontend)

```
document.addEventListener('DOMContentLoaded',
function() {
var currentDate = new Date();
var formattedDate = currentDate.toISOString().split('T')[0];
document.getElementById('leave-start-date').value = formattedDate;
});
```

```
document.addEventListener('DOMContentLoaded',
function() {
var startDateInput = document.getElementById('leave-start-
date');    var endDateInput =
document.getElementById('leave-end-date');
    function updateMinEndDate() {
endDateInput.min = startDateInput.value};
```

```

startDateInput.addEventListener('input', updateMinEndDate);
updateMinEndDate();
});

```

(Backend)

```

const firebaseConfig = {      apiKey:
    "AIzaSyDm_6dfAbs3dnW_NCew6CpzDEYeTkD1EpE",      authDomain:
    "hostelleaveformnew.firebaseio.com",      databaseURL:
    "https://hostelleaveformnew-default-rtdb.firebaseio.com",
    projectId: "hostelleaveformnew",      storageBucket:
    "hostelleaveformnew.appspot.com", messagingSenderId:
    "880538258215",      appId:
    "1:880538258215:web:ad65152ad187fccd057702"
};

```

```

firebase.initializeApp(firebaseConfig);

```

```

var hostelleaveformDB =
firebase.database().ref("hostelleaveform");

```

```

document.getElementById("hostelleaveform").addEventListener("submit",
submitForm);

```

```

function submitForm(e) {

e.preventDefault();
    var name = getElementVal("name");
    var emailid = getElementVal("emailid");
    var msgContent =
getElementVal("msgContent");

    saveMessages(name, emailid, msgContent);
    document.querySelector(".alert").style.display = "block";
    setTimeout(() => {
        document.querySelector(".alert").style.display = "none";

    }, 3000);
}

```



```
document.getElementById("hostelleaveform").reset();

    }    const saveMessages = (name, emailid,
msgContent) => {var
newhostelleaveform=hostelleaveformDB.push();
    newhostelleaveform.set({
name: name,emailid:emailid,

msgContent: msgContent, });
```

